gpeyt1-26 · Flow-net seepage

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A concrete diversion weir on a silty sand foundation (k = 2.5e-04 cm/s, isotropic) has headwater 34 ft and tailwater 6 ft above the foundation surface. A properly drawn flow net beneath the weir has Nf = 4 flow channels and Nd = 14 equipotential drops. The weir is 450 ft long. The sheet-pile cutoff is 20 ft deep and the sand is 45 ft thick.

Most nearly, the total seepage under the weir is:

A. 3,098 ft³/day
B. 31,252 ft³/day
C. 2,551 ft³/day
D. 6 ft³/day

Given equations / data

> q = k H Nf/Nd per unit length; H = total head loss; 1 cm/s = 2,835 ft/day

Worked solution

1. k = 2.5e-04 cm/s × 2,835 = 0.709 ft/day
2. H = 34 − 6 = 28 ft
3. q = 0.709 × 28 × 4/14 = 5.67 ft³/day/ft
4. Q = 5.67 × 450 = 2,551 ft³/day

Final answer

2,551 ft³/day

Why the other values are wrong

- **3,098 ft³/day** — used headwater depth instead of head difference
- **31,252 ft³/day** — inverted Nf/Nd
- **6 ft³/day** — reported per-foot flow (forgot length)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.16.4 Flow Nets, p. 190; §3.16.1 Darcy's Law

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k, https://www.youtube.com/watch?v=ng3In5ayyaE, https://www.youtube.com/watch?v=_qeKY4icSPs (no creator values copied).

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